EXECUTIVE BUSINESS REPORT

Superstore Sales Analysis – Power BI Dashboard

1. Executive Summary

This project focuses on analyzing Superstore sales data using Microsoft Power BI. The objective is to transform raw sales data into meaningful business insights through data modeling, DAX calculations, time intelligence, and interactive dashboard visualizations.

The dashboard provides an overview of sales performance, order volume, customer count, monthly trends, product categories, and regional performance.

The analysis identifies Technology as the highest-sales category, West as the highest-sales region, and 2018 as the strongest year in terms of sales.

2. Project Objectives

- Develop a structured data model using Power BI.
- Create a Calendar table and establish relationships.
- Build DAX measures for business performance analysis.
- Perform Month-over-Month and Year-over-Year analysis.
- Design an interactive executive-level dashboard.
- Generate actionable business insights and recommendations.

3. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

1. Total Sales: Approximately 2.26M
2. Total Orders: Distinct count of Order ID.
3. Total Customers: Distinct count of Customer ID.

These indicators help evaluate overall business performance and customer activity.

4. Year-wise Sales Analysis

Year	Total Sales
2015	479.86K
2016	459.44K
2017	600.19K
2018	722.05K

Key Findings:

- Sales decreased slightly in 2016 compared to 2015.
- Sales increased significantly in 2017.
- 2018 recorded the highest annual sales of 722.05K.
- Overall sales increased by approximately 50.5% from 2015 to 2018.

This indicates a positive sales growth trend over the analyzed period.

5. Category-wise Sales Analysis

The dashboard compares sales across three product categories:

- Furniture
- Office Supplies
- Technology

Key Finding:

Technology recorded the highest sales among all three categories.

Business Insight:

The business should continue monitoring Technology product demand and ensure adequate inventory availability to support sales performance.

6. Region-wise Sales Analysis

The dashboard compares sales across four regions:

- East
- West
- Central
- South

Key Finding:

The West region recorded the highest sales.

Business Insight:

The company can study the West region's sales performance and identify strategies that may help improve sales in other regions.

7. Time Intelligence Analysis

A dedicated Calendar table was created in Power BI and connected to the Superstore transaction table using Order Date.

The following DAX measures were developed:

1. Total Sales
2. Total Orders
3. Total Customers
4. Previous Month Sales
5. MoM Growth %
6. Previous Year Sales
7. YoY Growth %

These calculations support time-based performance comparisons and help identify changes in sales over different periods.

Year and Month slicers were also added to make the dashboard interactive.

8. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Focus on maintaining strong sales performance in the Technology category.
2. Analyze the West region's performance to identify successful sales strategies.
3. Monitor monthly and yearly sales trends to identify seasonal patterns.
4. Use KPI indicators to track sales, orders, and customer activity.
5. Apply time intelligence measures to support better business planning.
6. Conduct further profitability analysis using validated profit and cost data before making margin-related decisions.

9. Conclusion

The Superstore Sales Analysis project demonstrates how Power BI can transform raw business data into meaningful insights.

Through data modeling, DAX measures, time intelligence, and interactive visualizations, the dashboard provides a clear understanding of sales performance across different years, product categories, and geographical regions.

The findings highlight 2018 as the strongest sales year, Technology as the leading product category, and West as the highest-performing region.

This analysis can support data-driven decision-making, sales planning, and business performance monitoring